Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The acid present in red ants is:

(A) $HCOOH$

(B) C_6H_5COOH

(C) CH_3COOH

(D) CH_3COONa

Q.2

Corrosive sublimate $(HgCl_2)$ can be used to distinguish between:

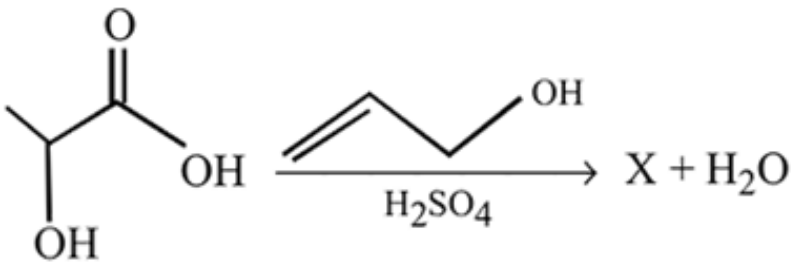
(A) formaldehyde and propanone

(B) formic acid and acetic acid

(C) acetaldehyde and butanone

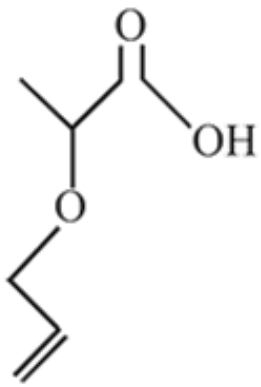
(D) all of the above

Q.3



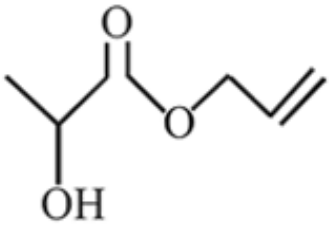
Product (X) of the reaction is-

(A)

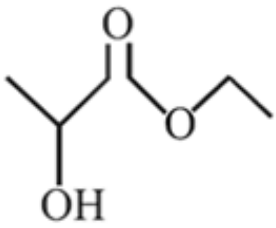


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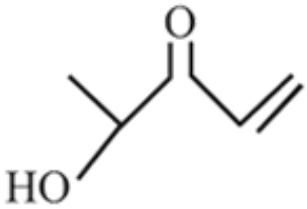
(B)



(C)



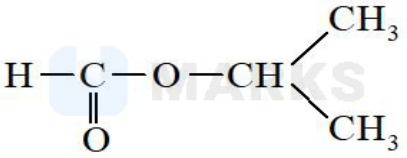
(D)



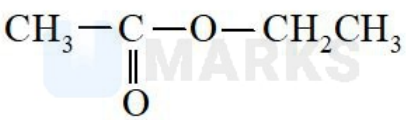
Q.4

An ester $A(C_4H_8O_2)$, on treatment with excess methyl magnesium chloride followed by acidification, gives an alcohol B as the sole organic product. Alcohol B, on oxidation with NaOCl followed by acidification, gives acetic acid. Ester A is:

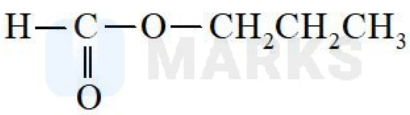
(A)



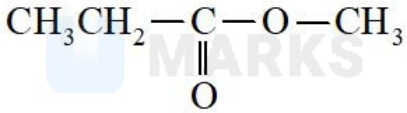
(B)



(C)



(D)



Q.5

Ethyl acetate is obtained when methylmagnesium iodide reacts with

(A) ethyl formate

(B) ethyl chloroformate

(C) acetyl chloride

(D) carbon dioxide

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Q.6

Benzoic acid may be converted to ethyl benzoate by reaction with :

(A) Sodium ethoxide

(B) Ethyl chloride

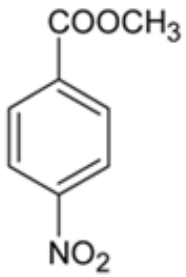
(C) Dry $HCl - C_2H_5OH$

(D) Ethanol

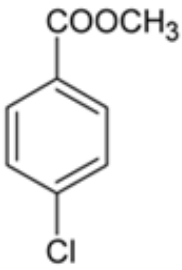
Q.7

The ease of alkaline hydrolysis is the greatest for

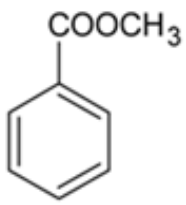
(A)



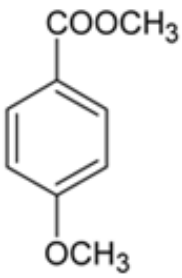
(B)



(C)



(D)



Q.8

Acetic acid differs from formic acid in that

(A) acetic acid is stable to heat

(B) formic acid is stable to heat

(C) acetic acid acts as a reducing agent

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(D) acetic acid shows positive test with Tollens reagent

Q.9

The reaction of formic acid with concentarted H_2SO_4 gives

(A) CH_3COOH

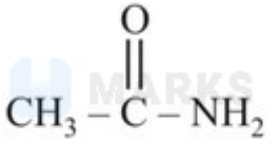
(B) $CO_2 + H_2O$

(C) $CO + H_2O$

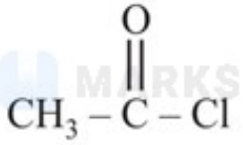
(D) $HCHO$

Q.10

Arrange reactivity of given compounds in decreasing order for hydrolysis reaction? < br /> < math> < mo>(< /mo> < mn>1 < /mn> < mo>) < /mo> < /math>



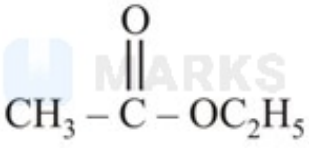
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< br /> < math> < mo>(< /mo> < mn>4 < /mn> < mo>) < /mo> < /math>



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Select the correct answer from the codes given below:

(A) 1, 2, 3, 4

(B) 2, 3, 4, 1

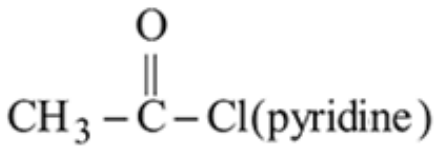
(C) 2, 3, 1, 4

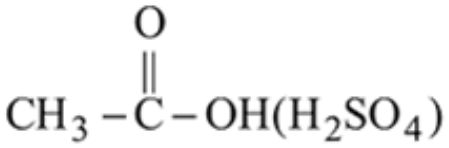
(D) 1, 4, 2, 3

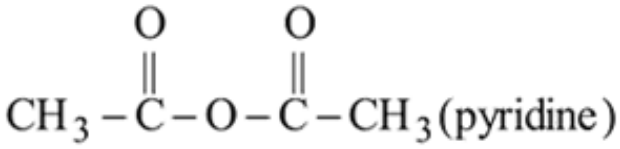
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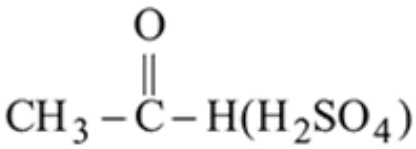
Q.11

Which of the following will not give butyl acetate when treated with 1 - butanol?

(A) 

(B) 

(C) 

(D) 

Q.12

The order of decreasing ease of reaction with ammonia is

(A) anhydrides, esters, ethers

(B) anhydrides, ethers, esters

(C) ethers, anhydrides, esters

(D) esters, ethers, anhydrides

Q.13

Which one of the following compounds is prepared in the laboratory from benzene by substitution reaction

(A) Glyoxal

(B) Cyclohexane

(C) Acetophenone

(D) Hexabromo cyclohexane

Q.14

Acetaldehyde reacts with semicarbazide and forms semicarbazone. Its structure is

(A) $CH_3CH = NNHCON = CHCH_3$

(B) $CH_3CH = NNHCONH_2$

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(C) $CH_3CH=N-\underset{\underset{OH}{|}}{N}-CONH_2$

(D) $CH_3CH=N-CONHNH_2$

Q.15

$CH_3CH_2Cl \xrightarrow{NaCN} X \xrightarrow{Ni/H_2} Y \xrightarrow[\text{anhydride}]{\text{Acetic}} Z$

Z in the above reaction sequence is

(A) $CH_3CH_2CH_2NHCOCH_3$

(B) $CH_3CH_2CH_2NH_2$

(C) $CH_3CH_2CH_2CONHCH_3$

(D) $CH_3CH_2CH_2CONHCOCH_3$

Answers & Solutions

Q.1 Answer:

$HCOOH$

Solution:

Formic acid is found in the bodies of real ants.

This acid is produced as a defensive mechanism for the ants.

Option A is correct.

Q.2 Answer:

formic acid and acetic acid

Solution:

Option B is the correct answer.

Formic acid gives white precipitate with $HgCl_2$ but acitic acid does not react.

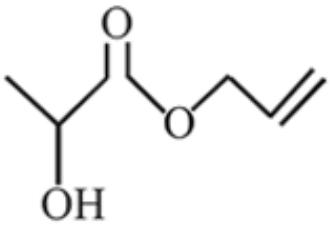
$HCO_2H + 2HgCl_2 \longrightarrow \underset{\text{white precipitate}}{Hg_2Cl_2\downarrow} + 2HCl + CO_2\uparrow$

$CH_3COOH \xrightarrow{HgCl_2}$ **No precipitate of** Hg_2Cl_2

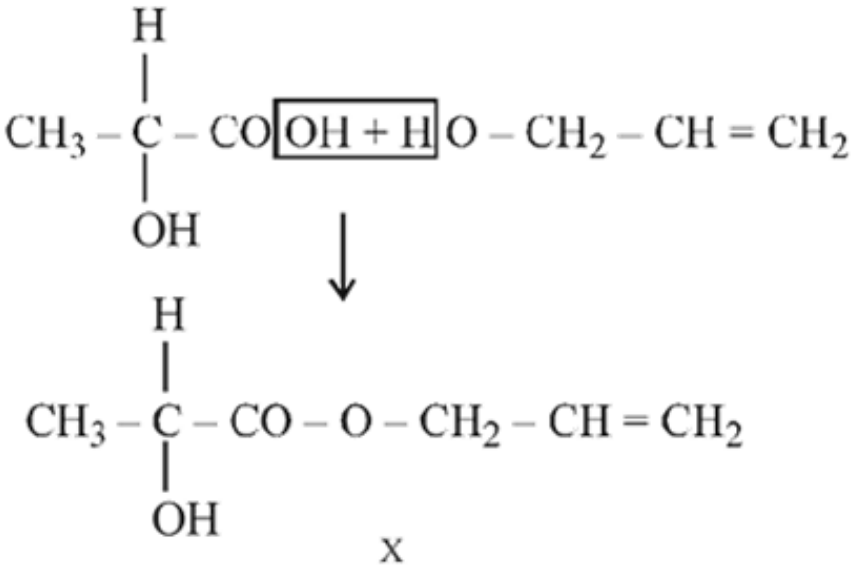
Formic acid acts as a reducing agent.

Q.3 Answer:

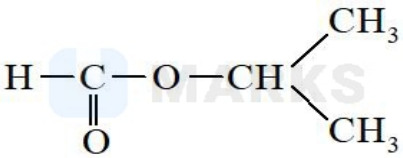
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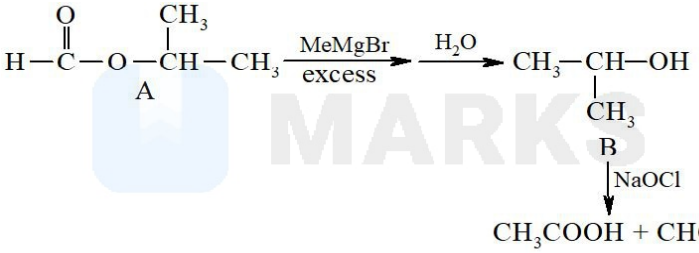
Solution:



Q.4 Answer:



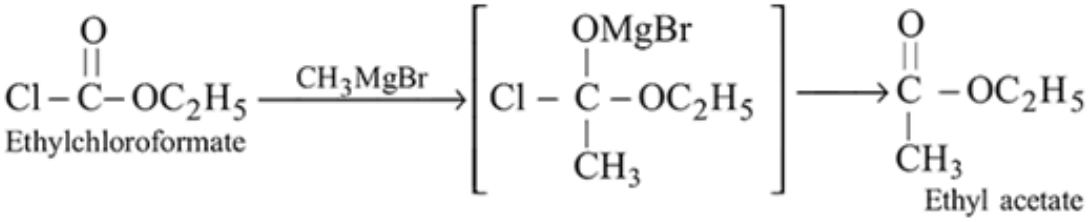
Solution:



Q.5 Answer:

ethyl chloroformate

Solution:

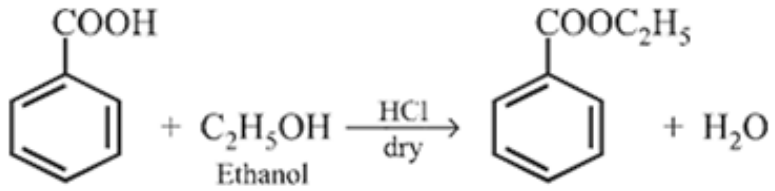


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Q.6 Answer:

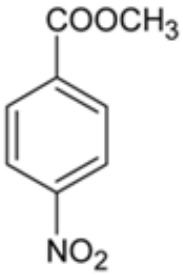
Dry $HCl - C_2H_5OH$

Solution:

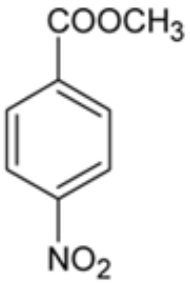


This process is known as esterification.

Q.7 Answer:



Solution:



It has the most electron deficiency on carbonyl carbon.

Q.8 Answer:

acetic acid is stable to heat

Solution:

Acetic acid differs from formic acid in that acetic acid is stable to heat.

Q.9 Answer:

$CO + H_2O$

Solution:

Formic acid undergoes dehydration when treated with conc. H_2SO_4

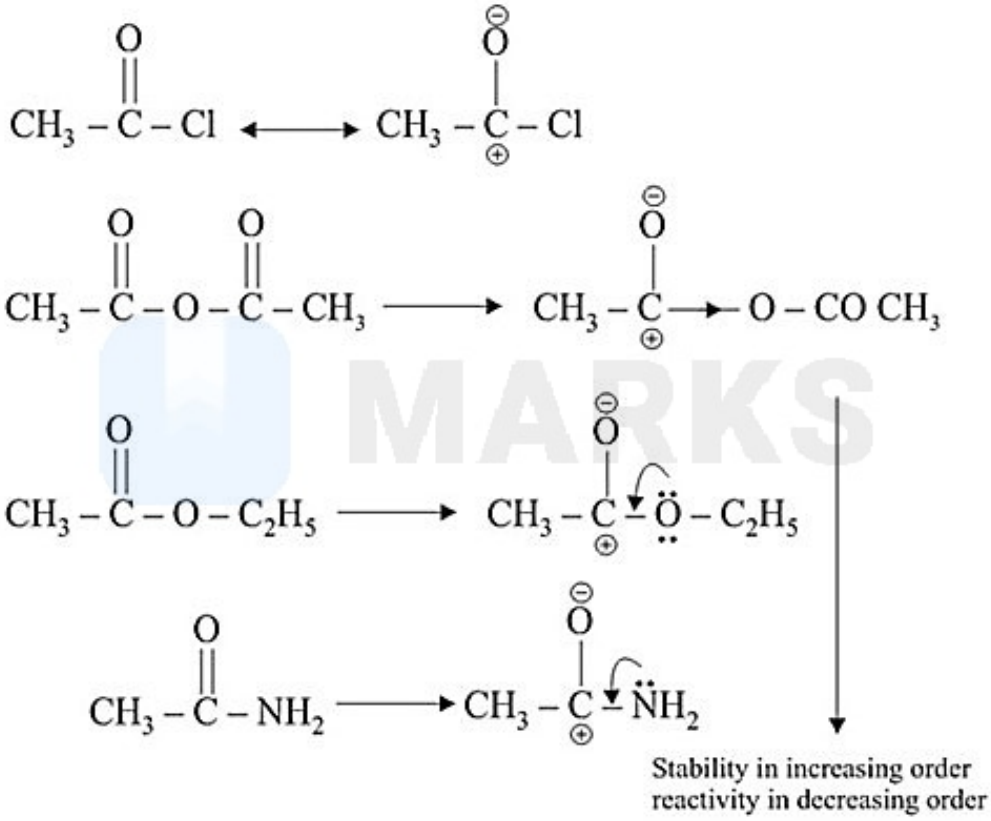
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Q.10 Answer:

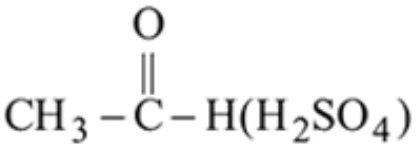
2, 3, 4, 1

Solution:

Here the decreasing order of hydrolysis is as follows < br />



Q.11 Answer:



Solution:

Acid chlorides, anhydrides, acids give ester with alcohol

$$ROH + R'COCl \longrightarrow ROCOR' + HCl$$
$$ROH + (R'CO)_2O \longrightarrow ROCOR' + RCOOH$$
$$ROH + R'COOH \longrightarrow ROCOR' + H_2O$$

Q.12 Answer:

anhydrides, esters, ethers

Solution:

Weaker the base, better the leaving group and hence more is the reactivity of the compound. The relative basic character of the four groups is

$$NH_2^- > OR^- > OCOCH_3^- > Cl^-$$

Thus the relative leaving ability of the four groups is

$$NH_2^- < OR^- < OCOCH_3^- < Cl^-$$

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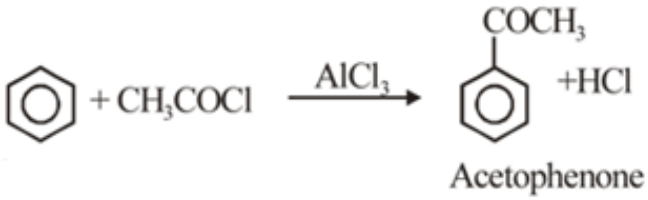
Hence the relative reactivity of the four acyl derivatives toward nucleophile is



Q.13 Answer:

Acetophenone

Solution:

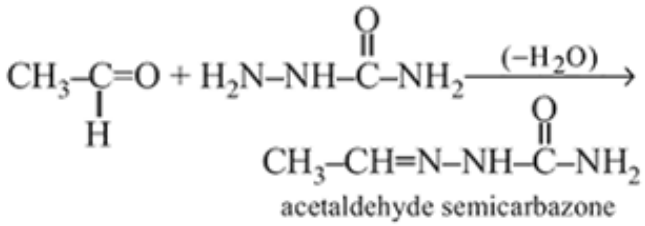


It is Friedel-Crafts reaction.

Q.14 Answer:

$CH_3CH = NNHCONH_2$

Solution:



Q.15 Answer:

$CH_3CH_2CH_2NHCOCH_3$

Solution:

$$CH_3CH_2Cl \xrightarrow{NaCN} \underset{(X)}{CH_3CH_2CN} \xrightarrow{Ni/H_2}$$

$$CH_3CH_2CH_2NH_2 \underset{(Y)}{(CH_3CO)_2O} \longrightarrow CH_3CH_2CH_2NHCOCH_3 + CH_3COOH$$

(Z)

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